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Exercise 2: BLE Localization

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1 Introduction

A nursing home for Alzheimer patients uses an IoT system to help caregivers locate dental prostheses that patients frequently misplace. Each denture is fitted with a small BLE beacon that periodically transmits advertising packets. BLE gateways installed throughout the facility receive these packets and estimate the approximate position of the denture based on RSSI measurements.

The current firmware follows a simple fixed policy: the beacon transmits one advertising event every 10 seconds, unconditionally, regardless of whether the denture has moved or whether any caregiver is actively searching for it. While straightforward to implement, this approach wastes energy during the many hours of the day in which the denture is resting in a stable, known location and frequent position updates carry no practical value.

The objective of this work is to redesign the beaconing logic and the sensing architecture of the beacon node in order to significantly extend battery lifetime while fully preserving the localization service quality during the periods when caregivers actually need to find the prosthesis.

2 Beaconing Logic

The proposed firmware replaces the fixed-rate policy with an event-driven adaptive strategy based on three operating states.

In the **MOVING** state, the accelerometer detects ongoing motion, indicating that the denture is likely being worn or physically handled. In this condition the denture is effectively co-located with the patient and does not require active localization. The node transmits one advertising event every 60 seconds, sufficient to confirm that the device is operational without drawing significant current from the battery.

When no motion is detected for $T_{\text{idle}} = 120\,\text{s}$, the system transitions to the **ALERT** state. This threshold was chosen as a deliberate compromise: it is long enough to absorb short periods of involuntary immobility, such as the patient sitting still while eating or watching television, which would otherwise generate spurious transitions to the ALERT state and waste energy unnecessarily. At the same time, it is short enough to ensure that the high-rate beaconing phase begins promptly after the denture is actually removed, since the act of placing the denture down is typically followed by a sustained period of immobility that well exceeds two minutes.

The ALERT state is the critical moment: the denture has just been placed somewhere and its position needs to be captured accurately before the patient walks away. The node therefore transmits one advertising event every 5 seconds for a duration of $T_{\text{alert}} = 5$ minutes, allowing the gateways to collect enough RSSI samples to compute a reliable position estimate.

After T_{alert} has elapsed with no further motion, the system enters the **STABLE** state. The denture is resting in a fixed, known location and frequent updates carry no additional information. The node transmits one advertising event every 5 minutes as a simple heartbeat, allowing the back-end to confirm that the device is still alive and to maintain the last known position.

Between any two consecutive transmissions, regardless of the current state, both the MCU and the BLE radio are fully powered down, reducing their combined current draw to well below $1\,\mu\text{A}$. The only component kept continuously energised is the MEMS accelerometer, which runs

autonomously in hardware wake-up mode. In this mode, the accelerometer internally compares the measured acceleration against a programmable threshold without any involvement of the MCU. When the threshold is exceeded, which indicates that the denture has started moving, the accelerometer asserts a dedicated interrupt output pin, which is physically connected to a wake-up pin of the MCU. This causes the MCU to resume execution within microseconds and unconditionally transition to the MOVING state. The key advantage of this architecture is that it is entirely event-driven: the MCU never needs to wake up periodically to poll the accelerometer and therefore remains in deep sleep for as long as no motion occurs, minimising energy consumption during the long periods in which the denture is either being worn or resting in a stable location.

3 Additional Sensors

The proposed solution adds a single additional sensor to the beacon node: a **low-power MEMS accelerometer**. The accelerometer serves as the sole source of context information for the adaptive beaconing logic described in Section 2, detecting the transition between motion and stillness that drives the state machine.

During the sensor selection phase, a humidity sensor was also considered as a candidate. The rationale was straightforward: the oral cavity maintains a relative humidity close to 100%, while the ambient humidity of a typical indoor environment is in the range of 40–60%. A humidity sensor could therefore provide a direct and unambiguous indication of whether the denture is being worn or not, without relying on motion inference.

However, this approach presents a fundamental incompatibility with the energy-saving objective of this work. Humidity sensors do not expose any interrupt or threshold detection capability, meaning that the only way to obtain a reading is to explicitly power on the sensor, trigger a measurement, and read the result. This forces the MCU to wake up periodically at a fixed rate regardless of whether the state of the device has changed, introducing a constant and unavoidable energy overhead that undermines the event-driven architecture described in Section 2.

The MEMS accelerometer, by contrast, performs threshold detection autonomously in hardware and signals state changes via an interrupt line, allowing the MCU to remain in deep sleep indefinitely until something physically changes. For this reason the accelerometer was selected as the sensing element for the proposed solution.

4 Improved Firmware Pseudocode

Listing 1: *Improved firmware pseudocode.*

```
// Constants
T_MOVING    = 60 s
T_ALERT     = 5 s
T_STABLE    = 300 s
T_ALERT_DUR = 300 s   // duration of ALERT state (5 minutes)
T_IDLE      = 120 s   // stillness threshold before MOVING -> ALERT

// Global variables
state          = MOVING
alert_start    = 0
```

```
last_motion    = 0

// Interrupt handler (called by accelerometer hardware interrupt)
function on_motion_detected():
    last_motion = now()
    state = MOVING

// Initialization
function setup():
    accelerometer.set_threshold(MOTION_THRESHOLD)
    accelerometer.enable_interrupt(on_motion_detected)
    accelerometer.start(mode = WAKE_UP)
    state = MOVING

// Main loop
function loop():
    send_BLE_beacon()

    if state == MOVING:
        if (now() - last_motion) > T_IDLE:
            state = ALERT
            alert_start = now()
        sleep(T_MOVING)

    else if state == ALERT:
        if (now() - alert_start) > T_ALERT_DUR:
            state = STABLE
        sleep(T_ALERT)

    else:  // STABLE
        sleep(T_STABLE)

// Entry point
function main():
    setup()
    loop forever:
        loop()
```

5 Commercial Components

The proposed solution requires two main hardware components: a BLE SoC and a MEMS accelerometer, powered by a coin cell battery. Several commercial options are available for each component.

For the BLE SoC, suitable candidates include:

- **Nordic Semiconductor nRF52805 / nRF52810** a small-footprint BLE 5.x SoCs with a typical TX current of approximately 5 mA at 0 dBm and a deep-sleep (System OFF) current of only 0.4 μA. They support wake-up via GPIO interrupt and are widely used in coin-cell beacon designs.
- **Nordic Semiconductor nRF52840** — a more powerful variant with additional memory and USB support. Slightly higher quiescent current but same interrupt-driven wake-up architecture.

- **STMicroelectronics BLUENRG-M2** — a single-chip BLE SoC with an integrated ARM Cortex-M0 core. Particularly compact and suitable for miniaturised devices such as a dental prosthesis beacon.

For the MEMS accelerometer, suitable candidates include:

- **Analog Devices ADXL362** — ultra-low-power 3-axis accelerometer with a measurement current of 1.8 μA at 100 Hz and a motion-activated wake-up mode drawing only 270 nA. Programmable activity and inactivity thresholds with a dedicated interrupt pin.
- **STMicroelectronics LIS2DH12** — comparable functionality with 1.8 μA in low-power mode and integrated activity/inactivity interrupt support. Widely adopted in IoT beacon designs.

For the energy source, a **CR2032 coin cell** is the natural choice, with a nominal capacity of 220 mAh at 3.0 V, corresponding to a total energy budget of approximately 2376 J.

For this work, the reference implementation assumes the nRF52810 paired with the ADXL362, as this combination minimises quiescent current while fully supporting the event-driven architecture described in Section 2.

6 Energy Saving Estimate

6.1 Assumptions

The following assumptions are used throughout the analysis.

On the hardware side, each BLE advertising event requires the CPU and radio to be active simultaneously, drawing approximately 6.8 mA for a total duration of 1.5 ms across three advertising channels, resulting in an energy cost of $E_{\text{beacon}} \approx 30.6\,\mu\text{J}$ per event. Between transmissions, the system draws a quiescent current of 1.77 μA (nRF52810 in System ON with RAM and RTC active, plus ADXL362 in wake-up mode), corresponding to a sleep power of $P_{\text{sleep}} \approx 5.31\,\mu\text{W}$. The power source is a CR2032 coin cell with a total energy budget of approximately 2376 J.

On the behavioural side, the denture is assumed to be worn for 12 hours per day, during which the system remains in MOVING state and transmits one beacon every 60 seconds. The remaining 12 hours are distributed across three removal events, each lasting approximately 4 hours. During the first 5 minutes of each removal event the system is in ALERT state and transmits one beacon every 5 seconds, for a total of 15 minutes per day. The remaining 705 minutes per day are spent in STABLE state, with one beacon every 5 minutes.

6.2 Baseline System

The original firmware transmits one beacon every 10 seconds unconditionally:

$$N_{\text{baseline}} = \frac{86400}{10} = 8640 \text{ beacon/day} \tag{1}$$

$$E_{\text{baseline}} = 8640 \times 30.6\,\mu\text{J} \;+\; 5.31\,\mu\text{W} \times 86400\,\text{s} = 264.4\,\text{mJ} + 459.1\,\text{mJ} = 723.5\,\text{mJ/day} \tag{2}$$

6.3 Improved System

The total number of beacon events per day in the improved firmware is:

$$N_{\text{MOVING}} = \frac{12 \times 3600}{60} = 720 \tag{3}$$

$$N_{\text{ALERT}} = \frac{15 \times 60}{5} = 180 \tag{4}$$

$$N_{\text{STABLE}} = \frac{705 \times 60}{300} = 141 \tag{5}$$

$$N_{\text{improved}} = 720 + 180 + 141 = 1041 \text{ beacon/day} \tag{6}$$

$$E_{\text{improved}} = 1041 \times 30.6\,\mu\text{J} \;+\; 5.31\,\mu\text{W} \times 86400\,\text{s} = 31.9\,\text{mJ} + 459.1\,\text{mJ} = 491.0\,\text{mJ/day} \tag{7}$$

6.4 Results

Quantity	Baseline	Improved
Beacons / day	8640	1041
Radio energy [mJ/day]	264.4	31.9
Sleep energy [mJ/day]	459.1	459.1
Total energy [mJ/day]	723.5	491.0
Battery lifetime	∼9.0 yr	∼13.2 yr

Table 1: *Daily energy budget comparison between baseline and improved firmware.*

The proposed solution reduces the radio energy by approximately 88%, from 264.4 mJ/day to 31.9 mJ/day. Including the unchanged sleep current, the total daily energy consumption is reduced by approximately 32%, extending the theoretical battery lifetime from 9.0 to 13.2 years on a CR2032 coin cell.

The benefit grows further in more sedentary scenarios. If the denture is worn for only 4 hours per day, for instance during extended hospital stays or periods of reduced patient activity, the number of daily beacon events drops to approximately 573, reducing total energy consumption to roughly 476.6 mJ/day and extending battery lifetime to approximately 13.7 years, corresponding to a total saving of about 34% over the baseline.

7 Conclusion

The proposed solution replaces the fixed-rate beaconing policy of the original firmware with an event-driven adaptive strategy that ties the transmission rate to the actual motion state of the denture. By adding a single ultra-low-power MEMS accelerometer operating in hardware wake-up mode, the system is able to distinguish between three meaningful operational contexts, the denture being worn, having just been placed down, and resting in a stable known location, and adjust the beacon rate accordingly.

The result is a significant reduction in radio activity, from 8640 to 1041 beacon events per day under the assumed usage model, while actually improving localization accuracy during the critical moments immediately after the denture is removed. The total daily energy consumption is reduced by approximately 32%, extending the theoretical battery lifetime from 9.0 to 13.2

years on a CR2032 coin cell.

It should be noted that the sleep current of the BLE SoC dominates the energy budget in both systems, which limits the achievable saving for a given hardware platform. Nonetheless, the proposed architecture demonstrates that a simple motion-aware beaconing logic, implemented with minimal additional hardware, can meaningfully extend battery lifetime without compromising the localization service.